

Q1.

A particle, P , of mass M is released from rest and moves along a horizontal straight line through a point O . When P is at a displacement of x metres from O , moving with a speed $v \text{ ms}^{-1}$, a force of magnitude $|8Mx|$ acts on the particle directed towards O . A resistive force, of magnitude $4Mv$, also acts on P .

- (a) Initially P is held at rest at a displacement of 1 metre from O . Describe completely the motion of P after it is released.

Fully justify your answer.

(8)

- (b) It is decided to alter the resistive force so that the motion of P is critically damped.

Determine the magnitude of the resistive force that will produce critically damped motion.

(4)

(Total 12 marks)

Q2.

A particle P , of mass m kg, moves in a straight horizontal line. At time t seconds, the displacement of P from a fixed point O on the line is x metres, and P is moving with velocity $\dot{x} \text{ m s}^{-1}$. Throughout the motion, two horizontal forces act on P : a force of magnitude $4mn^2|x|$ newtons directed towards O , and a resistance force of magnitude $2mk|\dot{x}|$ newtons, where n and k are positive constants.

- (a) Show that $\ddot{x} + 2k\dot{x} + 4n^2x = 0$.

(3)

- (b) In one case, $k = n$. When $t = 0$, $x = a$ and $\dot{x} = 0$.

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- (i) Show that $x = e^{-nt} \left(a \cos \sqrt{3}nt + \left(\frac{\sqrt{3}a}{3} \right) \sin \sqrt{3}nt \right)$.

(7)

- (ii) Show that P passes through O when $\tan \sqrt{3} nt = -\sqrt{3}$.

(3)

- (c) In a different case, $k = 2n$.

- (i) Find a general solution for x at time t seconds.

(3)

- (ii) Hence state the type of damping which occurs.

(1)

(Total 17 marks)

Q3.

A railway truck, of mass m , is travelling in a straight line along a horizontal track. At time $t = 0$, the truck strikes one end of a buffer which is fixed at its other end. The buffer may be modelled as a light spring of natural length a and modulus of elasticity $am n^2$, where n is a positive constant. At time t , the compression of the buffer is x .

- (a) In a simple model of the motion, the only force affecting the truck during this motion is the thrust from the buffer.
- (i) Show that, while the truck is in contact with the buffer, the truck performs simple harmonic motion. (2)
- (ii) Find, in terms of n , the period of this motion. (1)
- (b) In a more realistic model, the motion of the truck is affected by a resistance force of magnitude mkv , where v is the speed of the truck and k is a positive constant.
- (i) Show that, while the buffer is being compressed, x satisfies the equation
- $$\ddot{x} + K\dot{x} + n^2x = 0$$
- (ii) At time $t = 0$, the truck is travelling with speed U . Given that $k = \frac{5n}{2}$ find x in terms of n , U and t . (6)
- (iii) By means of a sketch, or otherwise, explain whether the type of damping is light, critical or heavy. (2)
- (Total 14 marks)

Q4.

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A particle P , of mass m , is suspended from a fixed point O by a light elastic string, of natural length a and modulus of elasticity $5mn^2a$, where n is a positive constant. When the particle hangs in equilibrium vertically below O , the extension of the string is $\frac{g}{5n^2}$. At time $t = 0$, the particle is projected vertically downwards from this equilibrium position with speed U .

During its subsequent motion, when P is moving with speed v , it experiences a resistance force of magnitude $2mnv$. The displacement of P below its equilibrium position at time t is x .

- (a) Show that, whilst the string is taut, x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 5n^2x = 0$$

(5)

- (b) Find x in terms of U and t .

(7)

- (c) State whether the nature of the damping caused by the resistance force is light, critical or heavy.

(1)

- (d) Find, in terms of U and π , the speed of P when it first returns to its equilibrium position.

(4)

(Total 17 marks)

Q5.

The ends of a light, uniform elastic string are fixed to two points, A and B , a distance $9a$ apart on a smooth, horizontal plane. The string is of natural length $6a$ and modulus of elasticity $4mn^2a$, where n is a constant.

A particle of mass m is attached to the string at P , where $AP = 6a$. The natural length of AP is $4a$ and the natural length of BP is $2a$. In this position, the particle is in equilibrium.



The particle is moved a distance $\frac{1}{2}a$ towards B and then released from rest at time $t = 0$. The displacement of the particle from its equilibrium position at time t is x .

Hence initially $x = +\frac{1}{2}a$.

The motion of the particle is resisted by a force of magnitude $2mnv$, where v is the speed of the particle at time t .

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- (a) Show that x satisfies

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 3n^2x = 0$$

(7)

- (b) Given that $n = 1$, find x in terms of a and t .

(8)

(Total 15 marks)

Q6.

A spring, of natural length $4a$ and modulus λ , has one end attached to a fixed support A , and a particle P of mass m is attached to its other end. Another spring, of natural length $2a$ and modulus $4mg$, has one end attached to P and the other end attached to a fixed support B , which is situated at a distance of $10a$ vertically below A . The system is in equilibrium in a vertical line with the upper spring stretched to a length of $7a$ and the lower

spring stretched to a length of $3a$ as shown in the diagram.



- (a) Show that $\lambda = 4mg$.

(4)

- (b) At time $t = 0$, the particle is lowered to a distance $\frac{a}{2}$ below its equilibrium position and released from rest. The subsequent motion of P is subject to a resistance of magnitude $\frac{1}{5}mkv$, where $k^2 = \frac{6g}{a}$ and v is the speed of the particle at time t .

- (i) Given that x is the downward displacement of P from its equilibrium position at time t , show that

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

(6)

- (ii) Hence find x in terms of a , k and t .

(9)

(Total 19 marks)

Q7.

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The motion of a particle, which moves along a straight line, is governed by the differential equation

$$2 \frac{d^2x}{dt^2} + 6 \frac{dx}{dt} + 9x = 0$$

where x is the displacement of the particle from the origin at time t .

Initially the particle is at rest at the point where $x = a$.

- (a) Find x in terms of a and t .

(9)

- (b) Is the damping of the motion of the particle light, critical or heavy? Give a reason for your answer.

(2)

(Total 11 marks)